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SIMATS ENGINEERING
ASSESSMENT TOOL I

36

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO5:
Measure network performance using key metrics with simulation tools
(BL4,BL5)

Assessment Tool Weightage: 40%

QUESTIONS: 3 **Total Marks:30**

Annexure A

Constraint-Based Problem Solving

1. A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability

2. A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.

3. A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Problem & Constraints	20%	Clearly identifies all constraints and fully understands the problem requirements	Identifies most constraints with minor gaps in understanding	Identifies some constraints but misses key aspects	Shows little or no understanding of constraints or problem
Problem Decomposition	15%	Breaks problem into logical, well-structured sub-problems effectively	Breaks problem into sub-parts with some logical flow	Attempts decomposition but lacks clarity or structure	No clear decomposition of the problem
Application of Constraints in Solution	20%	Applies all constraints correctly and consistently in solution design	Applies most constraints correctly with minor errors	Applies some constraints but with noticeable errors	Fails to apply constraints appropriately
Logical Reasoning & Strategy	15%	Uses strong, clear, and efficient reasoning strategies	Uses generally sound reasoning with minor gaps	Reasoning is partially correct but inconsistent	Reasoning is unclear or incorrect
Solution Accuracy & Feasibility	15%	Solution is fully correct, feasible, and optimized	Solution is mostly correct and feasible	Solution has errors or is partially feasible	Solution is incorrect or not feasible
Creativity & Innovation	5%	Demonstrates highly innovative and unique approach	Shows some creativity in approach	Limited creativity shown	No creativity, relies on basic or copied ideas
Clarity of Presentation	10%	Solution is clearly presented with excellent organization and explanation	Presentation is clear with minor issues	Presentation lacks clarity or organization	Poorly presented and difficult to understand
		9	10	9	8

COMPUTER NETWORKS CONSTRAINT-BASED PROBLEM SOLVING REPORT

Course Code: CSA07

Course Outcome Covered: CO5

Assessment Tool 1

(Expanded Edition)

Complete Question & Answer Document

Q1: A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability.

Answer:

To establish a reliable and cost-effective network in a remote rural location without high-speed wired internet, the most suitable connectivity solution is a hybrid Dual-WAN approach utilizing Low Earth Orbit (LEO) Satellite internet (such as Starlink) combined with a 4G/5G LTE cellular connection. LEO satellite provides the necessary high bandwidth and low latency required for cloud-based applications and video conferencing, while the LTE connection serves as a cost-effective failover to guarantee uninterrupted business operations.

For the networking devices, the branch should be equipped with an SD-WAN (Software-Defined Wide Area Network) Edge Router. SD-WAN intelligently routes traffic, prioritizing real-time data like video conferencing and VoIP using strict Quality of Service (QoS) policies. It dynamically monitors link health (jitter, latency, packet loss) and seamlessly switches to the LTE network if the satellite link degrades due to heavy weather. Additionally, a Managed PoE (Power over Ethernet) Switch should be deployed to connect internal devices, IP phones, and power wireless access points without requiring additional electrical wiring, significantly keeping deployment costs low. For network administration, a cloud-managed platform is essential, allowing IT staff at headquarters to configure, monitor, and troubleshoot the rural network remotely without expensive site visits.

To ensure secure communication with headquarters, a Site-to-Site IPsec VPN (Virtual Private Network) utilizing AES-256 encryption must be configured on the SD-WAN router. This creates a secure, encrypted tunnel over the public internet, protecting sensitive corporate data from interception. A Stateful Packet Inspection (SPI) Firewall with Intrusion Prevention System (IPS) capabilities should also be integrated at the edge to block unauthorized incoming traffic and mitigate advanced cyber threats.

This recommended architecture strictly adheres to the provided constraints. By combining satellite and LTE, it overcomes the bandwidth limitations of traditional rural networks while maintaining high reliability through automated failover. The SD-WAN router optimizes the limited budget by avoiding expensive leased lines (like MPLS) and maximizing the efficiency of affordable wireless links. Finally, this setup offers excellent future scalability; as the branch grows, additional LTE modems or higher-tier satellite plans can be seamlessly integrated into the existing SD-WAN infrastructure without requiring a complete hardware overhaul.

Q2: A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.

Answer:

Designing a campus-wide wireless network for over 3,000 students with a limited number of Access Points (APs) requires highly strategic capacity planning and intelligent hardware selection to maximize coverage. The physical placement of the APs must follow a staggered, high-density deployment model. In high-traffic zones like libraries and massive lecture halls, Wi-Fi 6 (802.11ax) APs with directional antennas should be deployed to focus signals exactly where students gather. Wi-Fi 6 is specifically designed for high-density environments, handling multiple simultaneous connections much more efficiently through OFDMA technology. In hostels and corridors, omnidirectional APs should be placed in central hallways to maximize the coverage footprint with fewer physical devices.

To minimize signal interference, the network must primarily utilize the 5 GHz frequency band, which offers significantly more non-overlapping channels than the older 2.4 GHz band. This reduces co-channel interference and provides the wider bandwidth necessary for rigorous academic research, software downloads, and multimedia streaming. The network controller will implement active band-steering to force capable devices onto the 5 GHz spectrum, leaving the 2.4 GHz band strictly for older legacy devices. Additionally, seamless roaming protocols

(802.11r/k/v) must be enabled so that student devices can transition between APs smoothly without dropping connections as they walk across campus.

For security, the network must implement WPA3-Enterprise authentication combined with an 802.1X RADIUS server. This ensures that every student logs in using their unique university credentials, allowing administrators to monitor individual usage, enforce bandwidth limits per user, and securely segment student traffic from faculty and administrative systems through dynamic VLAN assignment. A separate Captive Portal can be utilized for guest access, strictly isolating visitor traffic from internal academic resources.

This design perfectly balances the university's constraints. By using predictive RF heatmapping software before installation, the university can eliminate wireless dead zones and avoid over-purchasing hardware, rigorously adhering to the budget. Wi-Fi 6 and band-steering maximize the performance and lifespan of the limited APs. Finally, WPA3-Enterprise provides robust, enterprise-grade security with minimal ongoing operational costs, integrating flawlessly with the university's existing centralized student directory.

Q3: A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

Answer:

To meet the stringent security and compliance requirements of a financial institution without compromising network performance or vastly increasing the operational budget, a Zero-Trust Network Architecture based on logical network segmentation is absolutely critical. The core of this design relies on implementing Virtual Local Area Networks (VLANs) to segment the network logically rather than purchasing multiple expensive physical switches for each department. The network must be divided into strictly isolated VLANs: one for secure customer databases, one for general employee workstations, one for administrative management, and a Demilitarized Zone (DMZ) for external-facing online banking web servers.

To enforce strict access control boundaries between these segments, an internal Next-Generation Firewall (NGFW) should be deployed at the core of the network. The NGFW acts as a rigorous gateway, ensuring that a compromised employee workstation cannot directly communicate with the highly sensitive database VLAN. The NGFW will handle deep packet inspection (DPI) at wire speed, ensuring high throughput and negligible latency to preserve system performance. For continuous threat monitoring, an open-source or cost-effective Network

Intrusion Detection System (NIDS) should be deployed via port mirroring (SPAN) on the core switch. This allows the NIDS to passively analyze traffic for malicious anomalies, such as ransomware propagation or unauthorized data exfiltration, without introducing any inline network delay.

Secure access must be regulated through strict Identity and Access Management (IAM) protocols, including Multi-Factor Authentication (MFA) and Role-Based Access Control (RBAC). When employees attempt to access sensitive financial records, they must verify their identity using a hardware security key or authenticator app in addition to their password. RBAC ensures that employees only have access to the specific data necessary for their job roles, strictly adhering to the principle of least privilege. Furthermore, all internal data in transit must be encrypted using TLS 1.3, and data at rest utilizing AES-256 encryption, complemented by a Data Loss Prevention (DLP) system to monitor and prevent sensitive customer information from leaving the corporate network.

This architecture perfectly addresses the financial institution's constraints. It delivers maximum security and regulatory compliance by isolating sensitive data, enforcing encryption, and requiring MFA. High performance is maintained because the NIDS operates completely out-of-band, and the NGFW is hardware-accelerated for internal routing. Finally, budget constraints are met by utilizing logical VLAN segmentation and open-source NIDS rather than purchasing disparate hardware appliances for every department.